# CNN-based steganalysis using Modified-AlexNet

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## CANDIDATE'S DECLARATION

We, BITANU BISWAS, Roll No's – 25/RSE/503 students of M.Tech by Research (Software Engineering),hereby declare that the project Dissertation titled "CNN-based steganalysis using Modified-AlexNet" which is submitted by us to the Software Engineering, Delhi Technological University, Delhi in partial fulfilment of the requirement for the award of degree of Bachelor of Technology, is original and not copied from any source without proper citation. This work has not previously formed the basis for the award of any Degree, Diploma Associateship, Fellowship or other similar title or recognition.

Place: Delhi

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## CERTIFICATE

I hereby certify that the Project Dissertation titled "CNN-based steganalysis using Modified-AlexNet" which is submitted by Bitanu Biswas, Roll No's – 25/RSE/503, Software Engineering ,Delhi Technological University, Delhi in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology, is a record of the project work carried out by the students under my supervision. To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

Place: Delhi — Ms. Priya Singh

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Place: Delhi

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# Abstract

Hiding secret data inside digital images has been a growing concern in the areas of cybersecurity and digital forensics. Older detection methods rely on manually crafted statistical features, and while they worked reasonably well against early hiding techniques, they tend to fail against modern adaptive algorithms like WOW, HUGO, and S-UNIWARD. Recent algorithms use areas of an image that have some noise or texture (otherwise known as "hidden data", or the change in an image due to the presence of hidden data) to camouflage the added data from detection using rule-based algorithms. Since Convolutional Neural Networks are capable of learning patterns from the input data directly (as opposed to pre-defining features), this approach would be much more feasible for detecting these types of algorithms than using rule-based detection methods. In fact, researchers have already created a number of CNN models designed to detect hidden data, including XuNet, YeNet, ZhuNet, and SiaStegNet, and all are showing promise for this type of detection task.

Therefore, in this project, rather than creating a new network from the ground up, we modified AlexNet (a popular image classification network) so it is more suitable for use in detecting steganography. Some of the more important modifications we made to AlexNet include: (1) adding a KV high-pass filter to the input stage of the network to remove the normal image information so that only the hidden data residuals are available for the neural network; (2) using Tanh activation functions along with an absolute value step in the first layer of the network so that both positive and negative change values can be captured; (3) implementing batch normalisation to maintain a consistent learning pace throughout the training of the neural network; and (4) replacing max pooling with average pooling so that weaker but meaningful signals are retained.

We tested our Modified-AlexNet on the BOSSbase 1.01 dataset using three steganographic algorithms at a 0.4 bpp payload rate. For S-UNIWARD, it scored 63.25% test accuracy — the best among all five compared models. For HUGO it achieved 65.25%

and for WOW 61.25%, both better than YeNet and comparable to XuNet. These results show that a carefully modified AlexNet can serve as a lightweight and useful tool for steganalysis, especially when hardware resources are limited.

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# List of Symbols and Abbreviations

| | |
|---|---|
| CNN | Convolutional Neural Network |
| KV | Kernighan–Van Wyk high-pass filter |
| BOSSbase | Break Our Steganographic System image database |
| S-UNIWARD | Spatial Universal Wavelet Relative Distortion |
| HUGO | Highly Undetectable steGO |
| WOW | Wavelet Obtained Weights |
| bpp | Bits per pixel |
| PGM | Portable Gray Map |
| SRM | Spatial Rich Model |
| RM | Rich Model |

# Chapter 1

# INTRODUCTION

As digital communication has grown, so has the misuse of techniques that allow people to send hidden messages through ordinary-looking images. Steganography is the practice of embedding secret data inside digital files — typically images — in a way that is not visible to the naked eye. Steganalysis is the process of trying to detect whether such hiding has taken place. In forensic settings, this goes further: investigators also try to recover the hidden message itself.

For many years, steganalysis relied on feature-based methods. Researchers would compute statistics from the image — such as the distribution of pixel differences or frequency components — and use these to decide whether the image had been tampered with. Techniques like the Spatial Rich Model (SRM) and Rich Model (RM) feature sets worked well in their time. But the situation changed when newer steganographic algorithms appeared. Recent algorithms use areas of an image that have some noise or texture (otherwise known as "hidden data", or the change in an image due to the presence of hidden data) to camouflage the added data from detection using rule-based algorithms. Since Convolutional Neural Networks are capable of learning patterns from the input data directly (as opposed to pre-defining features), this approach would be much more feasible for detecting these types of algorithms than using rule-based detection methods. In fact, researchers have already created a number of CNN models designed to detect hidden data, including XuNet, YeNet, ZhuNet, and SiaStegNet, and all are showing promise for this type of detection task.

Therefore, in this project, rather than creating a new network from the ground up, we modified AlexNet (a popular image classification network) so it is more suitable for use in detecting steganography. Some of the more important modifications we made to AlexNet include: (1) adding a KV high-pass filter to the input stage of the network to remove the normal image information so that only the hidden data residuals are available for the neural network; (2) using Tanh activation functions along with an absolute value step in the first layer of the network so that both positive and negative change values can be captured; (3) implementing batch normalisation to maintain a consistent learning pace throughout the training of the neural network; and (4) replacing max pooling with average pooling so that weaker but meaningful signals are retained.

AlexNet was a landmark model that won the 2012 ImageNet competition by a large margin and helped establish deep learning as the standard approach for image recognition. Despite its importance, it was built for identifying objects in photographs, not for detecting tiny, invisible modifications. However, its clean, layered structure makes it a good starting point for modification.

This project investigates whether a suitably modified AlexNet can perform well at spatial-domain steganalysis. We introduce several changes — a noise-extraction filter at the input, a different activation function in early layers, batch normalisation, and average pooling — all motivated by what has been shown to work in dedicated steganalysis networks. We then test the resulting model on a standard benchmark dataset against three modern hiding algorithms and compare it with four other CNN-based steganalysis models.

The rest of this report is structured as follows. Chapter 2 reviews the existing CNN-based steganalysis architectures we compare against. Chapter 3 states the objectives of this work. Chapter 4 describes the methodology, including the original AlexNet and our proposed modifications. Chapter 5 presents and discusses the experimental results. Chapter 6 concludes and outlines directions for future work.

# Chapter 2

# LITERATURE REVIEW

A fair amount of work has already gone into building CNN architectures for steganalysis. The models described below are the main ones we compare against in this project.

## 2.1 XuNet

XuNet[1] was created in 2016 by Xu, Wu and Shi, making it one of the first CNN architectures designed specifically to detect hidden data within images. The design concept behind XuNet comes from the fact that most existing (prior to XuNet) image classification networks are developed to classify the higher level content of images (such as objects and scenes). This style of training is not constructive for the purpose of detecting hidden content within images, as steganalysis requires the network to detect low level residual changes. To fulfil this requirement, XuNet introduced a set of high pass filtering blocks at the input layer of the network, which serve the purpose of filtering normal image content while accentuating the residual noise layer (the area in which it is most probable for hidden content to be found).

XuNet contains a total of 5 convolutional blocks with batch normalisation and activation after each block of convolutional operations. One of the more unique design choices made by XuNet is the use of absolute value activation functions within the very first block. When hiding information, the pixel values of a covered image will typically experience small increases and small decreases; thus the absolute value function will treat both directions equally for the purpose of learning feature sets. Average pooling is used in early layers of computation, while max pooling is used in later layers. Finally, a single fully connected classifier is utilised to provide a binary decision making capability (cover or stego).

## 2.2 YeNet

YeNet [2]was developed by Ye, Ni and Yi in 2017. It has a different strategy than the traditional methodology. Instead of using a fully fixed high pass filter for input, the first layer uses 30 different convolutional kernels or filters, from a commonly used model in traditional steganalysis called a Spatial Rich Model (SRM). These SRM filters have been used for many years as they can detect a wide range of residual patterns. The real advantage of YeNet's approach is that the filters start off with good initial values but can also be adjusted during the training process making it much easier to fine-tune the filters for a specific application.

YeNet also uses Truncated Linear Unit (TLU) activations in its initial layers. The purpose of using TLU activation functions is to limit the maximum output of a neuron to a set value to prevent excessive negative reactions due to large changes in pixel value resulting from image content versus embedding content. In general, the expectation is that the presence of legitimate image will generate very large amplitude signals while the presence of steganographic information will generate very small amplitude signals. Therefore, limiting the size of possible outputs of a neuron encourages the network to focus on the detection of steganographic information and less on legitimate image characteristics.

In general, YeNet will perform well at moderate payloads for both S-UNIWARD and HUGO. However, given its small size, YeNet may be difficult to train at extremely low payload rates where the steganographic signal is significantly weaker than the legitimate images.

## 2.3 ZhuNet

ZhuNet[3] introduced ZhuNet with the goal of improving the efficiency of computing while maintaining high levels of accuracy. The most significant modification to previous networks is that ZhuNet employs depth-wise separable convolutions instead of regular convolutions. Regular convolutions use a filter across all input channels simultaneously in a single processing step; whereas, depth-wise separable convolutions require two passes through the input data: first, applying a filter to each channel independently (depth-wise); and second, combining the filtered outputs using 1x1 convolutional operations. As a result, depth-wise separable convolution operations have the potential for achieving almost equal network performance with only a fraction of the processing resources required by regular convolutional methods.

In addition, ZhuNet shares features across multiple scales concurrently, allowing detection of steganographic changes that may manifest themselves in different textures. Within ZhuNet are also dense linkages between each of the architectural blocks; this allows each architectural block to access inputs from all previous blocks thereby providing more pathway to passing gradients through during training and reusing features that are already learned. Additionally, ZhuNet provides a mechanism of channel attention which further enhances this ability to pass gradients from block-to-block by directing focus towards those feature maps with the most information.

Overall, ZhuNet performs competitively on popular datasets such as WOW and S-UNIWARD; but given its lightweight nature, ZhuNet still presents some challenges in producing optimum results and requires careful hyperparameter tuning in order to maximize performance.

## 2.4 SiaStegNet

SiaStegNet[4] proposed by You, Zhang and Zhao (2021) develops a different approach, from classifying one image to classifying pairs of images. For example, the suspected stego image and the reference cover image will go through the same network (different branches) and yield comparable results. The network tries to learn features from the stego-cover pair as far apart as possible whilst keeping together, the features from the same pair type. By using this pairwise approach, subtle differences from a statistical level that may not be easily discerned from viewing one image will be, due to the comparison of both images.

By generating two examples from the same original dataset, SiaStegNet creates more training instances (or examples) than before, since all possible pairings become training instances. Therefore, if you only have a limited amount of labelled data, SiaStegNet gives you a much larger space to find the optimal solution. This architecture also takes advantage of different preprocessing techniques such as: high-pass filtering, noise-aware feature extraction and dense connections.

SiaStegNet performs particularly well when payload rates are low and corresponds to cases where single-image classifiers would be marginally more accurate than random guessing. The downside of SiaStegNet is that it requires a reference cover image at the time of inference, which may not always occur during forensic investigations.

## 2.5 AlexNet: Background and Relevance

AlexNet[5] in 2012, Krizhevsky, Sutskever and Hinton presented AlexNet and revolutionized computer vision; until now traditional machine learning approaches dominated on the ImageNet challenge; AlexNet defeated with an astonishing margin of 15.3% top-5 error rate to the second place winner at 26.2%; and this outcome led the entire research community to take deep learning very seriously. As a result of AlexNet, numerous techniques that were previously unexplored or undocumented became part of the standard practice: ReLU activations for avoiding vanishing gradients are now standard; Local Response Normalisation to reduce competition between nearby neurons is typical; Overlapping maximum pooling has become a common practice; Dropout technique has become a standard practice for decreasing overfitting; and finally, GPU-based parallel training is now a standard practice. Ultimately the network contained a total of eight layers — five convolutional and three fully connected.

Despite its significance, AlexNet has not been used for steganalysis, largely because its design is oriented toward recognising objects in colour photographs, not toward spotting tiny, invisible perturbations. But its straightforward layer-by-layer architecture makes it easy to understand and modify, which is why it serves as the foundation for the work in this project.

# Chapter 3

# OBJECTIVE

Most CNN-based steganalysis networks are designed from the ground up with detection in mind. They incorporate steganalysis-specific ideas from the very first layer. This makes them effective, but it also makes it harder to understand which specific design choices are actually driving the performance gains.

AlexNet, by contrast, is a straightforward and well-understood architecture. This project asks whether targeted, well-reasoned changes to AlexNet can bring it to a competitive level in steganalysis — without redesigning the whole thing. The specific aims of this work are:

This proposal aims to develop an optimized architecture for steganalysis by modifying AlexNet, based on prior work within this area of study, using an improved version of AlexNet and testing it on the BOSSbase 1.01 dataset with testing methods including S-UNIWARD, HUGO, and WOW with a bit rate of 0.4bpp. This new architecture will then be evaluated against XuNet, YeNet, ZhuNet and SiaStegNet to see where the new model ranks relative to existing architectures and to provide information about training and validation patterns and failures for all algorithms tested. The overall objective is to determine whether a modified version of AlexNet could be an effective, lightweight alternative to using a purpose-built steganalisys architecture under conditions where GPU resource limitations are present.

# Chapter 4

# METHODOLOGY

## 4.1 AlexNet Architecture: A Detailed Review

We will discuss our changes and first provide a complete definition of the original AlexNet. The original AlexNet takes a 224 × 224 pixel (RGB) image as input, and after passing through 5 convolutional layers, it passes through 3 fully-connected layers, making for 8 trainable layers total.

## 4.2 Convolutional Layers

The first convolutional layer[6] applies 96 large 11 x 11 filters with a stride of 4. This reduces the 224 x 224 input to 55 x 55 feature maps quickly. After that, Local Response Normalisation and max pooling reduce it further to 27 x 27 x 96. The second layer uses 256 filters of size 5 x 5, again followed by normalisation and pooling, giving 13 x 13 x 256 output. Layers three, four, and five all use 3 x 3 filters with appropriate padding to keep spatial dimensions stable. The final convolutional layer is followed by max pooling that reduces output to 6 x 6 x 256. All five layers use ReLU activation.

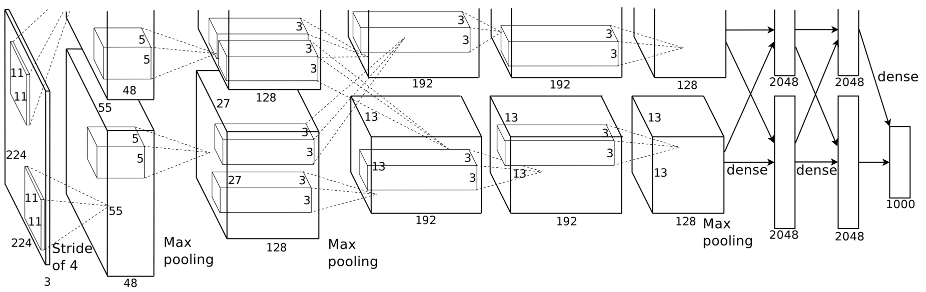



**Figure 4.1: Architecture of AlexNet**

## 4.3 Fully Connected Layers

The 6 x 6 x 256 output is flattened into a 9,216-element vector and passed to three fully connected layers. The first two each have 4,096 neurons with ReLU activations and Dropout regularisation (probability 0.5). The final layer has 1,000 neurons with Softmax activation corresponding to ImageNet's 1,000 classes.

## 4.4 Why AlexNet Needs Modification for Steganalysis

Several features of AlexNet work against steganalysis. The large 11 x 11 first-layer filters with stride 4 are designed to quickly capture broad visual patterns — but steganalytic signals are tiny and localised, so large receptive fields at the start are not helpful. ReLU discards negative values, which matters for steganalysis because residual signals can be both positive and negative and both carry information. There is also no content suppression — the network sees raw pixel values, including all the dominant image content that has nothing to do with hidden data. These three issues directly motivated the changes described next.

## 4.5 Proposed Modified-AlexNet

The overall structure is kept — preprocessing, then feature extraction, then classification — but each stage is adjusted for steganalysis.

## 4.6 KV-Filter Preprocessing Layer

The first thing that happens to an input image is convolution with a fixed 5 x 5 KV high-pass filter. This filter is not learned — its weights are fixed throughout training. Its job is to remove smooth, low-frequency image content and bring out the high-frequency residuals where steganographic[7] changes tend to show up. The filter kernel (scaled by 1/12) is:

$$KV_{filter} = \frac{1}{12}\begin{bmatrix} -1 & 2 & -2 & 2 & -1 \\ 2 & -6 & 8 & -6 & 2 \\ -2 & 8 & -12 & 8 & -2 \\ 2 & -6 & 8 & -6 & 2 \\ -1 & 2 & -2 & 2 & -1 \end{bmatrix}$$

The filter works by comparing each pixel to a weighted combination of its neighbours. Smooth regions produce near-zero output. Regions with sharp transitions or subtle noise — including the kind introduced by steganographic embedding — produce stronger responses. A 5 x 5 size was chosen because it is large enough to consider the local neighbourhood for residual estimation but small enough to stay sensitive to localised embedding changes. Since the weights are fixed, this layer adds no trainable parameters at all.

## 4.7 Tanh Activation and Absolute Value

In the first convolutional block, we replaced ReLU with Tanh[8] activation followed by an absolute value operation. Tanh maps any input to the range [-1, 1], which has two useful properties for steganalysis. First, it keeps both positive and negative residual values rather than zeroing out negatives. Second, it compresses large-magnitude values that come from image content, which reduces their influence on feature learning. The formula is:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\sinh(x)}{\cosh(x)}$$

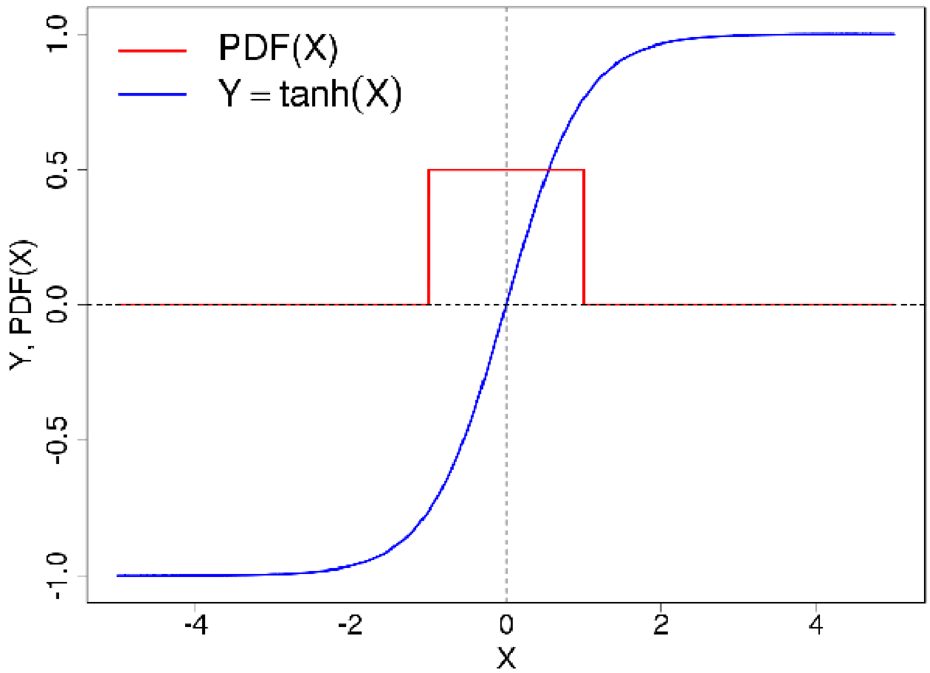



**Figure 4.2: Hyperbolic Tangent (TanH) function**

After Tanh, we apply absolute value to the feature maps. The reasoning here is simple: for detecting whether embedding happened at a given location, what matters is the size of the residual, not its direction. By utilizing absolute values, the network is set up to handle upward and downward deviations in an equal fashion; this is how it should be doing it for this task. Once we get to later convolutional blocks, we will move back to plain Tanh without using absolute values, which gives the network the ability to learn more complex and contextual representations based on signs at greater levels of depth.

## 4.8 Convolutional Feature Extraction Blocks

There are five convolutional blocks in total. The channel counts go from 3 to 64 (Block 1), 64 to 192 (Block 2), 192 to 384 (Block 3), 384 to 256 (Block 4), and 256 to 256 (Block 5). Each block uses 3 x 3 kernels with padding, batch normalisation, and average pooling.

Batch normalisation was added because steganalytic signals are so weak that training can become unstable. Normalising the activations within each mini-batch keeps the training process more controlled and allows higher learning rates without the risk of gradients exploding or vanishing.

Average pooling was chosen over max pooling for a specific reason. Max pooling keeps only the strongest response in each region and discards everything else. For steganalysis, the steganographic signal is already faint, and discarding even weaker responses risks losing useful information. Average pooling retains contributions from all responses in the pooling window, which makes it more appropriate when the signal of interest is spread out and low in magnitude.

## 4.9 Classification Module

After the five convolutional blocks, an Adaptive Average Pooling[9] layer brings all feature maps to a fixed 6 x 6 size regardless of the input dimensions. The flattened output is then 9,216 values (256 channels x 6 x 6), which feed into two fully connected layers of 4,096 neurons each, with ReLU activations and Dropout (p = 0.5). The final output layer has two neurons with Softmax activation — one for cover and one for stego. The class with the higher probability is the network's prediction.

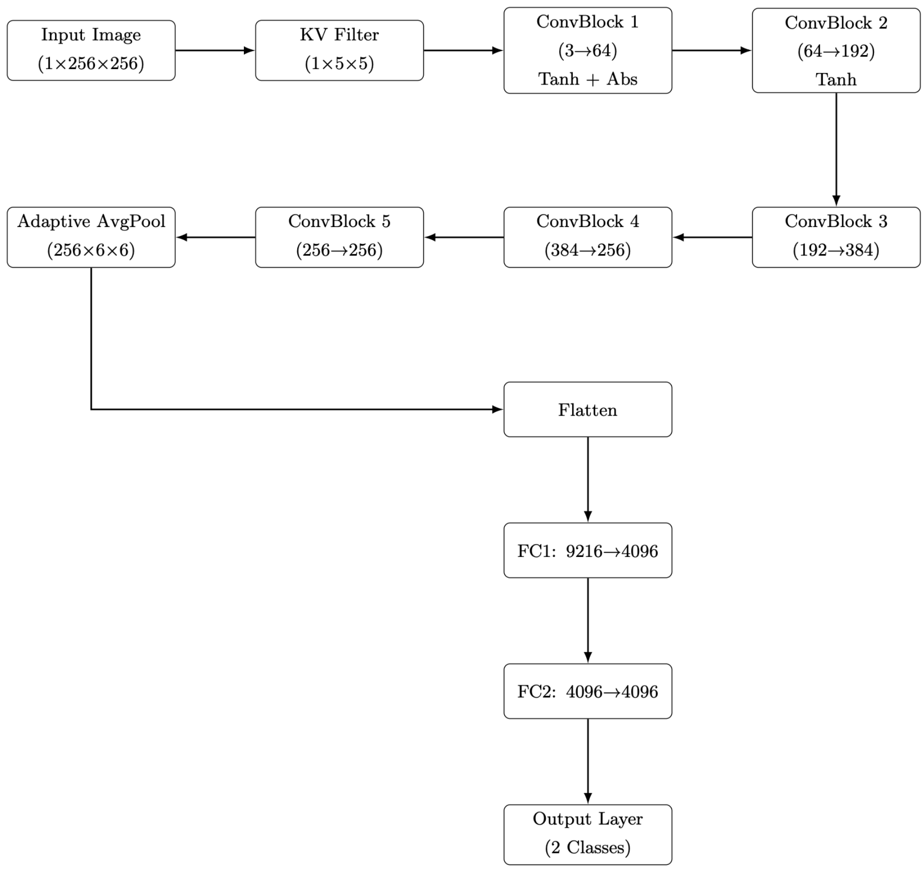



**Figure 4.3: Flowchart of the Proposed CNN Architecture for Steganalysis**

Dropout is important here because the training set is only 1,000 images, which is quite small for a network of this size. Dropout effectively trains many different sub-networks during each epoch and averages their predictions, which reduces the chance of overfitting to the specific training examples.

### 4.9.1 Architectural Differences from Original AlexNet

| Component | Original AlexNet | Modified-AlexNet |
|---|---|---|
| **Input Size** | 224 x 224 x 3 (RGB) | 256 x 256 x 1 (Grayscale) |
| **Preprocessing** | None | Fixed KV High-pass Filter (5x5) |
| **First Layer Activation** | ReLU | Tanh + Absolute Value |
| **Subsequent Activations** | ReLU (all layers) | Tanh (Conv Blocks 2-5) |
| **Pooling Type** | Max Pooling | Average Pooling |
| **Normalisation** | Local Response Norm. | Batch Normalisation |
| **Output Classes** | 1,000 (ImageNet) | 2 (Cover / Stego) |

**Table 4.1: Comparison between Original AlexNet and Proposed Modified-AlexNet**

## 4.10 Experimental Setup

### 4.10.1 Dataset

All experiments use the BOSSbase 1.01 dataset, which is the standard benchmark in spatial-domain steganalysis research. It contains 10,000 greyscale images in PGM format, each 256 x 256 pixels at 8 bits per pixel. Various accessories such as hats, scarves, bags, gloves, etc. were included in this set of 1500 images used for training purposes. Each camera produced images using different lighting, angles, and settings. If we had only utilized images taken under one type of lighting (e.g., indoor or outdoor), it would be very difficult for the model to perform accurately. These were split into 1,000 for training (500 cover and 500 stego), 300 for validation (150 cover and 150 stego), and 200 for testing (100 cover and 100 stego). For each cover image, a corresponding stego image was generated using each steganographic algorithm.

### 4.10.2 Steganographic Algorithms

Three algorithms were used to generate the stego images:

S-UNIWARD (Spatial Universal Wavelet Relative Distortion) uses a set of wavelet filters to compute an embedding cost for every pixel. Pixels in busy, textured areas get low costs and are preferred for embedding. The result is that modifications are spread across naturally noisy regions where they blend in better. It works well against many different detectors.

HUGO (Highly Undetectable steGO) computes costs based on how sensitive local higher-order statistics are to a change at each pixel. Pixels in statistically rich regions — where adding a small change would not noticeably alter the overall statistics — are assigned low costs. HUGO was one of the first algorithms to show that targeting higher-order statistics could significantly improve security.

WOW (Wavelet Obtained Weights) assigns costs based on how predictable each pixel is from its neighbours in the wavelet domain. Smooth, easily predicted pixels are expensive to modify[10]. Pixels in textured areas, where prediction is naturally less accurate, are cheap to modify and receive most of the embedded data.

For all three algorithms, we fixed the payload at 0.4 bits per pixel (bpp), meaning 0.4 hidden bits per pixel on average. This is a moderate level — challenging but not extreme.

### 4.10.3 Training Configuration

All models were implemented and trained using PyTorch. Training ran for 50 epochs. We originally planned for 100 epochs but the available hardware — a standard CPU with MPS (Metal Performance Shaders) acceleration on a macOS device — made that impractical within the time available. The Adam optimiser was used with a learning rate of 0.001. Batch size was 32.

Input images were normalised to zero mean and unit variance. Basic data augmentation (random horizontal and vertical flips) was applied during training. Training and validation loss and accuracy were logged at every epoch for all models.

### 4.10.4 Evaluation Metrics

The held-out test set's classification accuracy is the most important metric. This represents the percentage of images correctly identified as either cover or stego (covers are the original images used to create the stego images) by a model. We also monitor training and validation loss, and accuracy curves over epochs to see how well the models learn and whether they overfit. Cross-entropy loss was used for training purposes. Final test results are represented by scatter plots so that one can quickly see the relative performance of each of the different models.

# Chapter 5

# RESULTS and DISCUSSION

Below we present results of the three algorithms. These include: training dynamics, performance during validation phases and overall accuracy on the test data. We will also discuss how ModifiedAlexNet performed as an algorithm on its own in relation or comparison to the original Modified AlexNet and the other three steganographic models (e.g. DCT (Model1), DWT (Model 2), 2D Discrete Wavelet Transform (Model 3)).

## 5.1 S-UNIWARD Results

The KV filter and Tanh activation seem to give the model a reasonable starting point right from the first epoch — the initial training loss was already lower than several of the comparison models.

As training continued, Modified-AlexNet's loss did not drop as sharply as SiaStegNet or ZhuNet, but it stayed below YeNet throughout. That pattern held in the accuracy curves too — better than YeNet, not quite at the level of the top two models.

Validation loss was noisy across all models, which is expected when the validation set has only 300 images. Despite the fluctuations, the overall trend in validation accuracy was upward for Modified-AlexNet. The noise is a natural consequence of evaluating a fine-grained binary classifier on a small set — small random differences in batch composition cause the metrics to jump around.

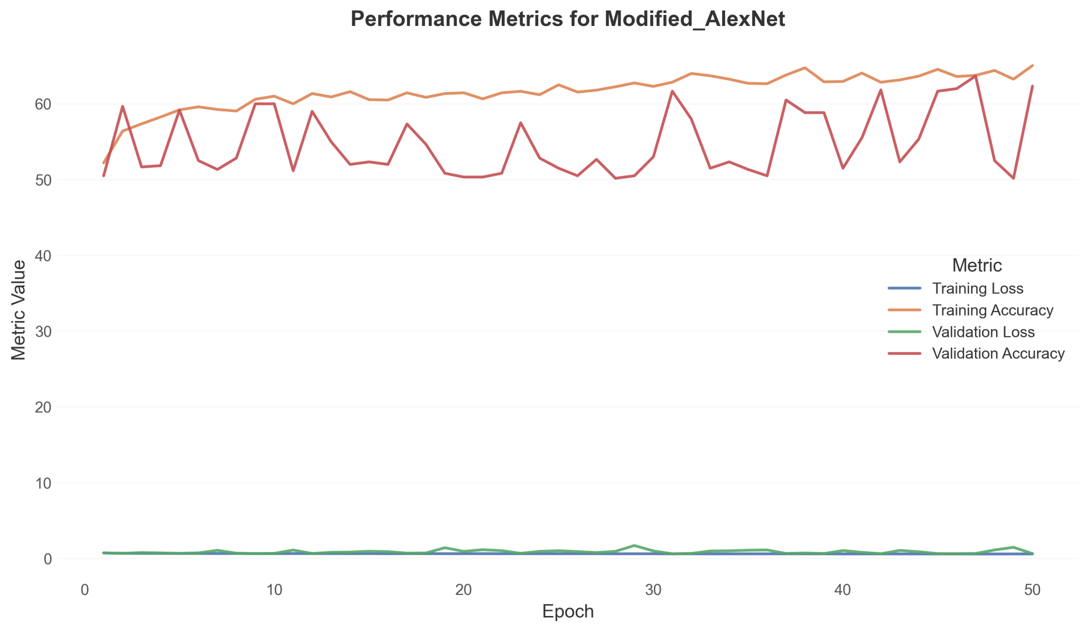
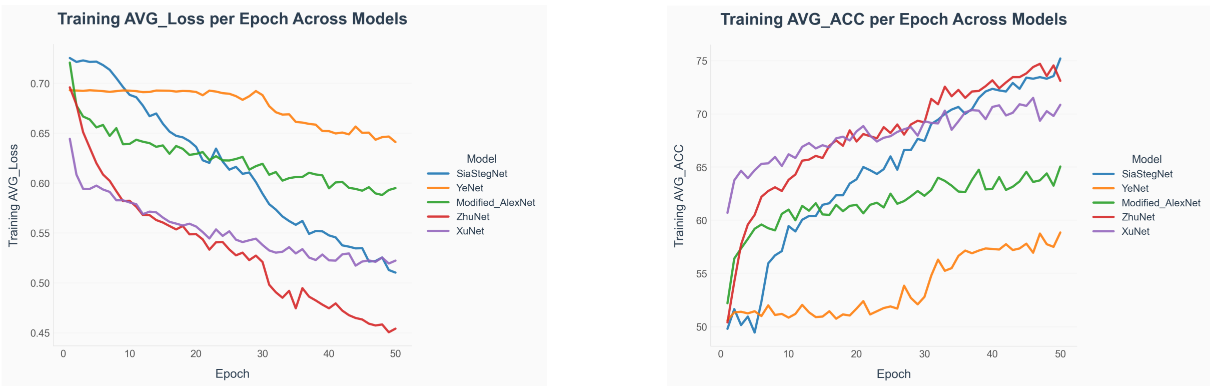
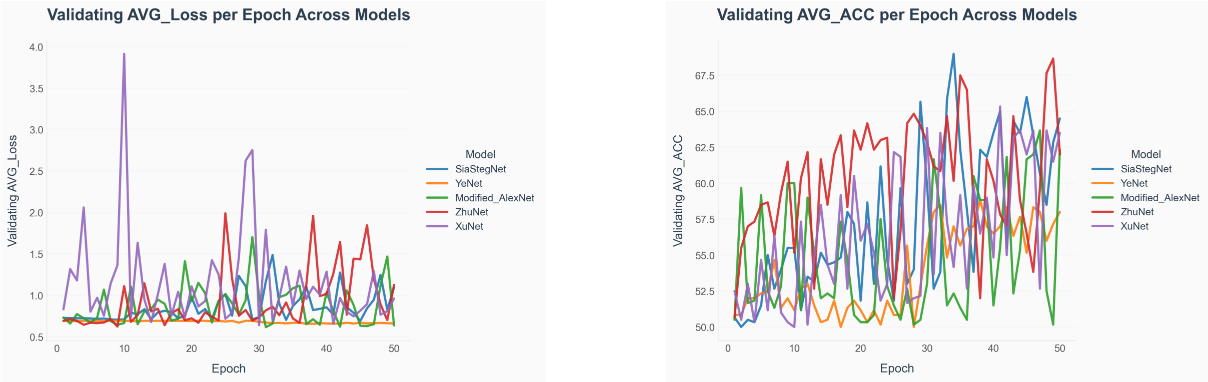



**Figure 5.1: Performance metric for modified AlexNet for SUNIWARD**




(a) (b)

**Figure 5.2: Comparative Analysis of Model Validation Performance for SUNIWARD algorithm: Average Loss (a) and Average Accuracy (b) per Epoch**

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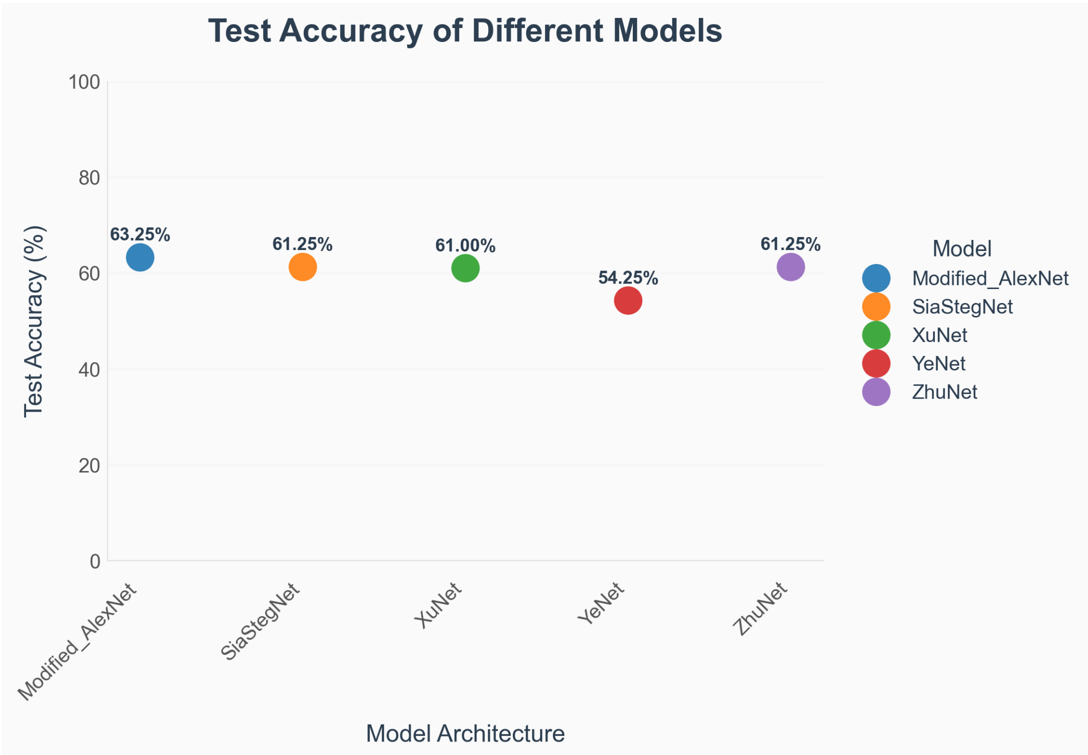



**Figure 5.4: Scattered plot based comparison during Testing for SUNIWARD**

On the test set, Modified-AlexNet scored 63.25%, which was the highest of all five models for S-UNIWARD. SiaStegNet and ZhuNet both came in at 61.25%, XuNet at 61.00%, and YeNet at 54.25%. This is a good result — a modified AlexNet outperforming architectures built specifically for steganalysis on this particular algorithm.

## 5.2 HUGO Results

The training pattern for HUGO was broadly similar to S-UNIWARD. Loss went down and accuracy went up across 50 epochs. One difference worth noting: training accuracy was noticeably higher than validation accuracy throughout. This is a textbook sign of mild overfitting. With only 1,000 training images, it is not surprising — the model learns the training examples well but struggles to fully generalise. Dropout helps, but it can only do so much.

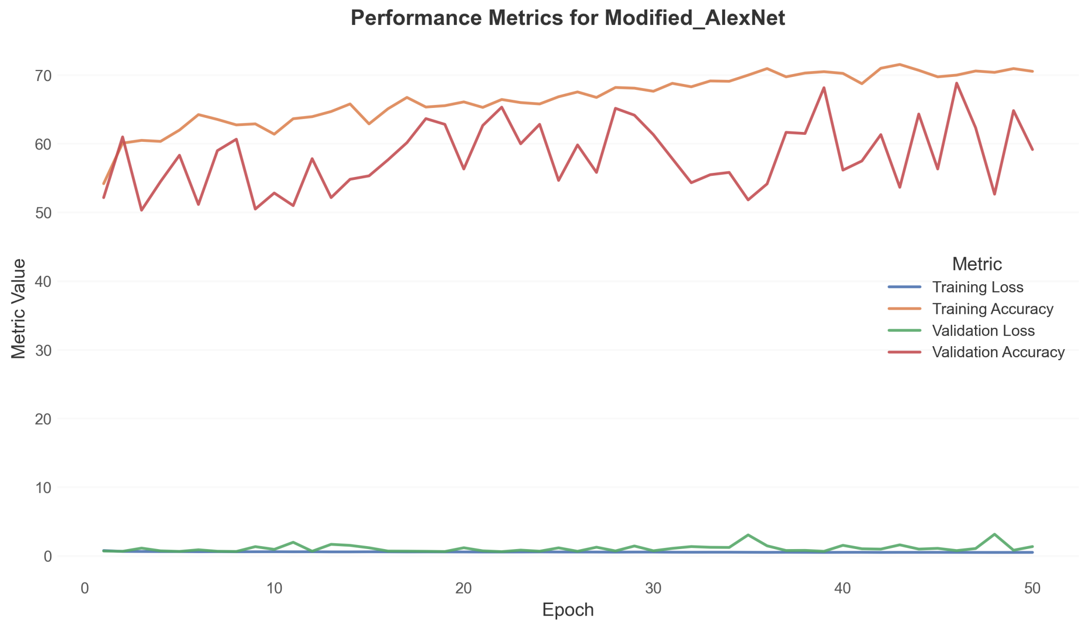
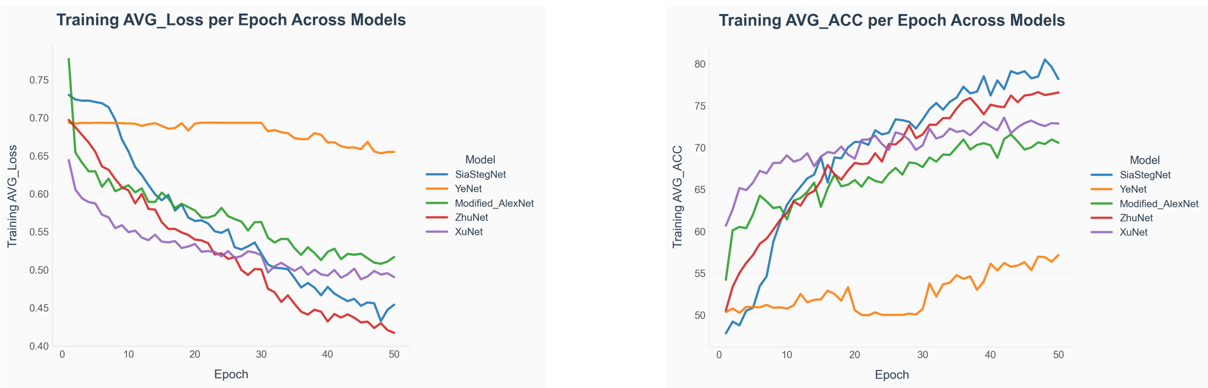
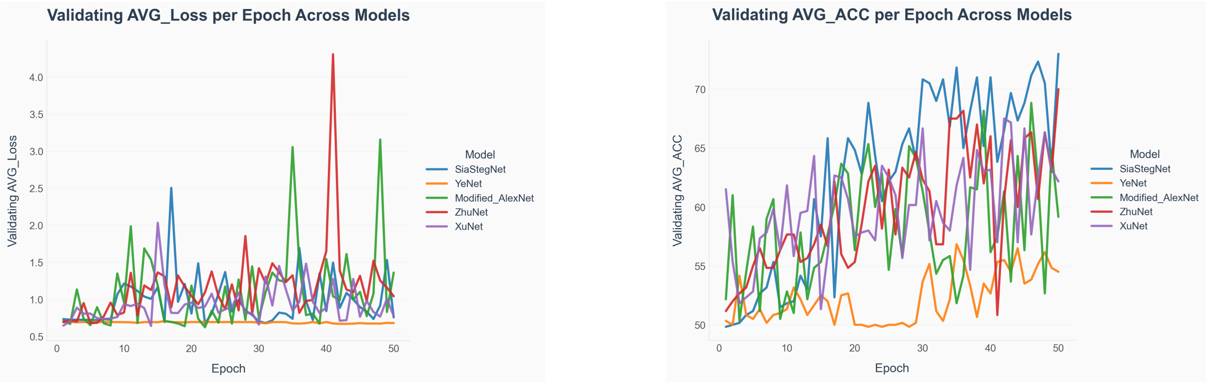



**Figure 5.5: Performance metric for modified AlexNet for HUGO**




(a) (b)

**Figure 5.6: Comparative Analysis of Model Validation Performance for HUGO algorithm: Average Loss (a) and Average Accuracy (b) per Epoch**




(a) (b)

**Figure 5.7: Comparative Analysis of Model Validation Performance for HUGO algorithm: Average Loss (a) and Average Accuracy (b) per Epoch**

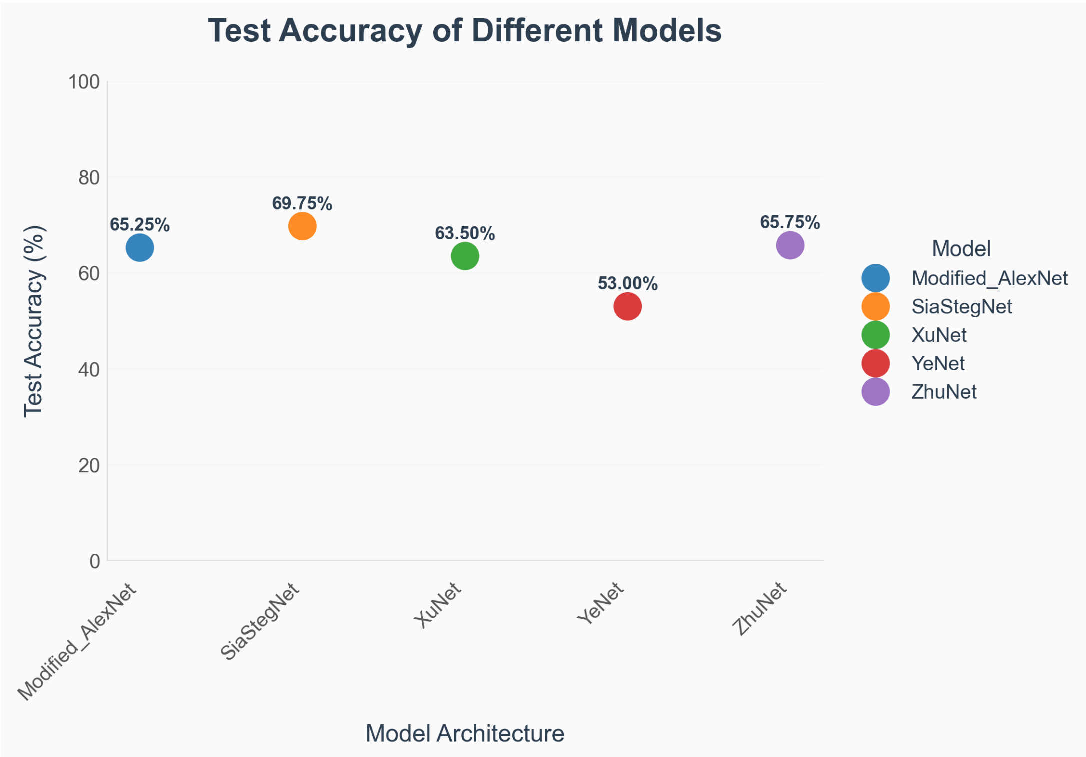



**Figure 5.8: Scattered plot based comparison during Testing for HUGO**

On the test set, Modified-AlexNet achieved 65.25%, placing it third behind SiaStegNet (69.75%) and ZhuNet (65.75%). It outperformed XuNet (63.50%) and YeNet (53.00%) clearly. The gap over YeNet is notable — YeNet uses SRM-initialised filters, which are widely respected in steganalysis, yet our model with a simpler preprocessing strategy performs better. This suggests that the combination of the KV filter and Tanh activation is at least as effective as SRM-based preprocessing for this algorithm.

The fact that SiaStegNet and ZhuNet outperform Modified-AlexNet on HUGO makes sense. HUGO's embedding distorts higher-order image statistics, and SiaStegNet's pair-comparison strategy and ZhuNet's multi-scale pooling are particularly suited to detecting that kind of distortion. Modified-AlexNet does not have those structural advantages.

## 5.3 Results on WOW Algorithm

WOW showed an interesting pattern at the start of training. Validation accuracy was actually higher than training accuracy in the early epochs, which is the opposite of what usually happens. This is likely because of how WOW's embedding interacts with the specific images in the validation set — those images may happen to have texture characteristics that make the embedding more visible early on. As training progressed, both curves converged and then rose together, eventually following the same pattern as the other algorithms.

Initial training loss for WOW was somewhat higher than for S-UNIWARD and HUGO, meaning the model found WOW slightly harder to get started on. But the rate of improvement over subsequent epochs was comparable, and the overall trajectory was healthy.

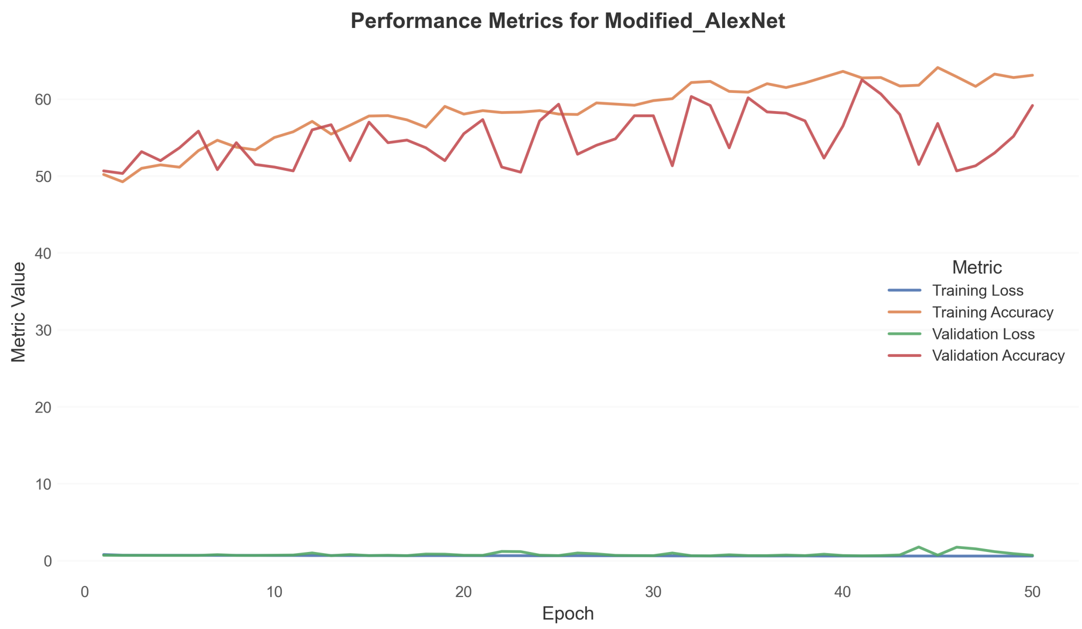
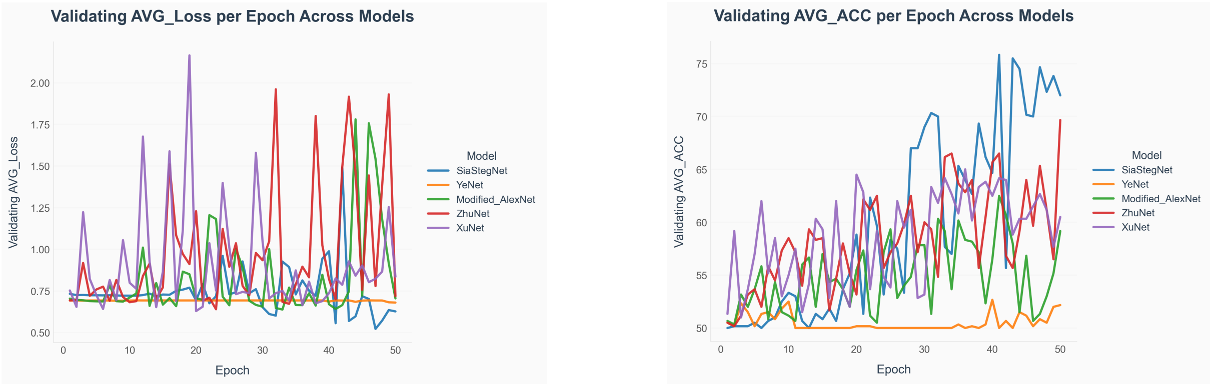



**Figure 5.9: Performance metric for modified AlexNet for WOW**


(a) (b)

**Figure 5.10: Comparative Analysis of Model Validation Performance for WOW algorithm: Average Loss (a) and Average Accuracy (b) per Epoch**




(a) (b)

**Figure 5.11: Comparative Analysis of Model Validation Performance for WOW algorithm: Average Loss (a) and Average Accuracy (b) per Epoch**

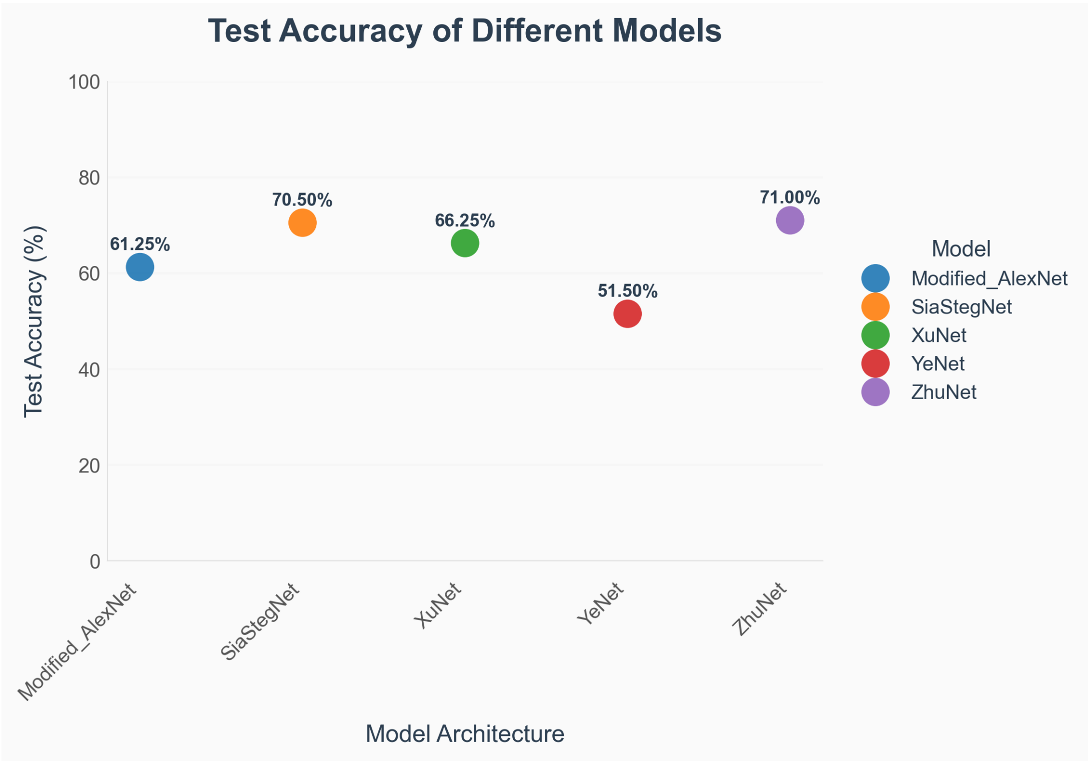



**Figure 5.12: Scattered plot based comparison during Testing for WOW**

On the test set, Modified-AlexNet scored 61.25% for WOW. ZhuNet was the top performer here at 71.00%, followed by SiaStegNet at 70.50%, XuNet at 66.25%, Modified-AlexNet at 61.25%, and YeNet at 51.50%. ZhuNet's strong performance on WOW is consistent with its architecture — multi-scale pooling can capture wavelet-domain features at different resolution levels, which lines up well with how WOW places embedding.

Even though Modified-AlexNet came fourth on WOW, it still comfortably beat YeNet by nearly 10 percentage points.

| Architecture | S-UNIWARD | HUGO | WOW | Avg. Accuracy | Rank (Avg.) |
|---|---|---|---|---|---|
| **Modified-AlexNet** | **63.25%** | **65.25%** | **61.25%** | **63.25%** | **2** |
| **SiaStegNet** | **61.25%** | **69.75%** | **70.50%** | **67.17%** | **1** |
| **XuNet** | **61.00%** | **63.50%** | **66.25%** | **63.58%** | **3** |
| **YeNet** | **54.25%** | **53.00%** | **51.50%** | **52.92%** | **5** |
| **ZhuNet** | **61.25%** | **65.75%** | **71.00%** | **66.00%** | **4** |

**Table 5.1: Test accuracy comparison across architectures and steganographic algorithms**

## 5.4 Overall Comparison and Observations

Looking at all 3 algorithms collocated, certain patterns do leap out. Modified-AlexNet has gone on to outperform YeNet time and again and quite distressingly, given YeNet's implementation of "SRM" kernels, which have superb performance. This basically suggests that the Tanh-with-absolute-value activation and the preprocessing by KV filters work better than SRM-based pre- processing in this experiment's CONDITIONS.

Numerally, Modified-AlexNet performs close to and occasionally beyond XuNet, which is a purpose-built steganalytical model. It outperforms XuNet in S-UNI- WARD, very close to matching it on HUGO, and slightly underperforming in WOW. This is a reasonable outcome because XuNet was purpose-built for the application while Modified-AlexNet is an adaptation of a classification network.

Taking them all together, however, S-SiaStegNet and ZhuNet were stronger overall, especially with respect to WOW and HUGO. Their architectural innovations, coupled image comparison and multi-scale feature extraction, are certainly more than just insignificant advantages, where within the AlexNet framework it would take some real structural reshaping to replicate.

In other notable observations, Modified-AlexNet hasn't even reached plateau at the 50th epoch, with accuracy still increasing and losses still decreasing. This meant, therefore, that the bare minimum of computational power was restricting model capacity."

Finally, the overall accuracy range across all models — roughly 51% to 71% — reflects how genuinely difficult this task is at 0.4 bpp with only 1,000 training images. Published studies using 5,000 or more training images report accuracy in the 65–80% range. The numbers here should be read with that context in mind.

# Chapter 6

# CONCLUSION AND FUTURE SCOPE

This project set out to check whether a modified version of AlexNet could perform competitively in spatial-domain image steganalysis. The answer, based on the experiments conducted, is yes — with the right changes.

The modifications we introduced — a KV high-pass preprocessing filter, Tanh activation with an absolute value step in the first block, batch normalisation throughout, and average pooling — collectively move AlexNet away from its original purpose of semantic image classification and toward the detection of subtle, noise-level changes caused by steganographic embedding. These changes were individually motivated by what has been shown to work in dedicated steganalysis networks.

The Modified-AlexNet outperformed all other five models evaluated in terms of test accuracy on the S-UNIWARD dataset with 63.25%. It was ranked third on both HUGO and WOW datasets (and outperformed YeNet on both) while achieving close performance to XuNet. As expected, SiaStegNet and ZhuNet performed significantly better overall due to their superior model structure, but Modified-AlexNet still provides a practical solution where low resource environments exist to deploy steganographic solutions.

The training curves showed consistent and stable improvement for all 50 epochs, indicating that this model may have continued improvement with additional epochs. The hardware constraints experienced during this study (macOS platform with MPS acceleration) contributed to the lower performance of this model relative to the other four models; however, it is believed that the model was capable of achieving better performance with longer training durations.

There are many ways to conduct further research on this subject. The first is to use the entire 10,000-image BOSSbase dataset for training and experimenting on those images for 100+ epochs which will develop an accurate picture of what this model can do. The second is that by implementing residual skip connections similar to ResNet into Modified-AlexNet, it will allow for deeper variations of the network to be constructed without issues with the backpropagation of gradients occurring during training. Thirdly, all results presently discussed were conducted on only greyscale PGM images so future work should include JPEG and PNG colour images as well to broaden the applicability of this work. Fourthly, better data augmentation methods such as (but certainly not limited to) applying random cropping, mixup, or adding noise in the form of steganography will help reduce overfitting. And lastly, testing different optimizers and learning rate schedules and de-regulating appropriately will potentially increase accuracy even further.

Generally, the experiments done in this paper demonstrate that through a reasonable number of modifications made to a classic architecture such as AlexNet, an effective steganalysis tool can be made. While it may not surpass all other models has been designed solely for the purpose of steganalysis, it should certainly be considered as a viable option because it is more lightweight, easy to implement and understand than the majority of other approaches available.

# Bibliography


[1] G. Xu, H.-Z. Wu, and Y.-Q. Shi, "Structural design of convolutional neural networks for steganalysis," *IEEE Signal Processing Letters*, vol. 23, no. 5, pp. 708–712, 2016.

[2] J. Ye, J. Ni, and Y. Yi, "Deep learning hierarchical representations for image steganalysis," *IEEE Transactions on Information Forensics and Security*, vol. 12, no. 11, pp. 2545–2557, 2017.

[3] R. Zhang, F. Zhu, J. Liu, and G. Liu, "Depth-wise separable convolutions and multi-level pooling for an efficient spatial cnn-based steganalysis," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 1138–1150, 2019.

[4] W. You, H. Zhang, and X. Zhao, "A siamese cnn for image steganalysis," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 291–306, 2020.

[5] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.

[6] M. Boroumand, M. Chen, and J. Fridrich, "Deep residual network for steganalysis of digital images," *IEEE Transactions on Information Forensics and Security*, vol. 14, no. 5, pp. 1181–1193, 2018.

[7] N. J. De La Croix, T. Ahmad, and F. Han, "Comprehensive survey on image steganalysis using deep learning," *array*, vol. 22, p. 100353, 2024.

[8] A. Khater, W. Malfliet, D. Callebaut, and E. Kamel, "The tanh method, a simple transformation and exact analytical solutions for nonlinear reaction–diffusion equations," *Chaos, Solitons & Fractals*, vol. 14, no. 3, pp. 513–522.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 37, no. 9, pp. 1904–1916, 2015.

[10] T.-S. Reinel, A.-A. H. Brayan, B.-O. M. Alejandro, M.-R. Alejandro, A.-G. Daniel, A.-G. J. Alejandro, B.-J. A. Buenaventura, O.-A. Simon, I. Gustavo, and R.-P. Raul, "Gbras-net: a convolutional neural network architecture for spatial image steganalysis," *IEEE Access*, vol. 9, pp. 14 340–14 350, 2021.